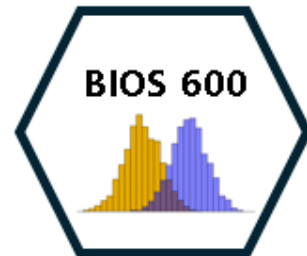


Lecture 15: Power and sample size

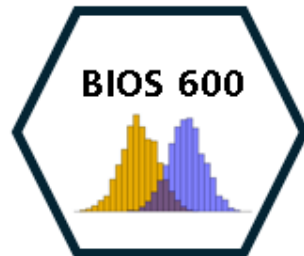
BIOS 600 - Fall 2025

UNC - Chapel Hill



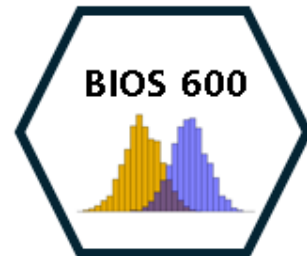
Announcements

- ▶ Lab 6 due Friday at 11:59pm
- ▶ AE 04 due Monday 10/28 at 11:59pm
- ▶ HW 6 will be assigned later today, due next Thursday at 11:59pm



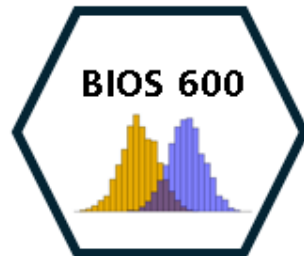
Overview

- ▶ Why calculate sample size and power?
- ▶ How to calculate sample size and power in R
- ▶ Application Exercise



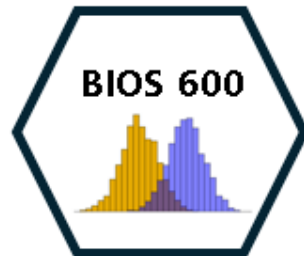
Very optional reading

- ▶ P&G Sections 10.4 and 10.6
- ▶ OI: No corresponding section



Why calculate sample size and power?

- ▶ To show that under certain required conditions, a hypothesis test has a **good chance of showing the anticipated difference**, if it really exists
- ▶ To be more confident that a null result is not simply a sample of excessive variability
- ▶ To show a funding agency that the study has a reasonable chance of reaching a useful result (leading them to **fund your grant!**)
- ▶ To show that necessary resources (human, animal, financial, time, etc.) will be minimized



Power

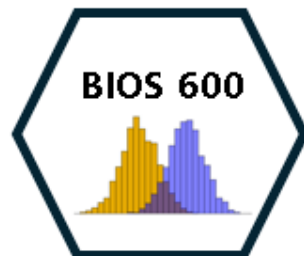
- ▶ Showing adequate statistical power is usually necessary in order to **get funding for research!**
- ▶ **Power** is the probability of rejecting the null hypothesis when it is false (i.e., of avoiding a Type II error)

$$\text{Power} = P(\text{reject } H_0 | H_0 \text{ is false})$$

and can be also thought of as the likelihood a planned study will detect a deviation from the null hypothesis if one really exists.

Power is a function of:

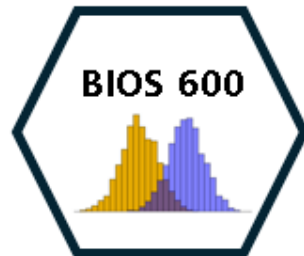
- ▶ Sample size n
- ▶ Deviation from the null one hopes to detect
- ▶ Standard deviation σ
- ▶ α , the Type I error rate



Power and Error Rates

- Recall: When we conduct a hypothesis test, there are **four possible outcomes** depending on the true state of the null hypothesis and the decision we make.

Decision	Truth: (H_0) True	Truth: (H_0) False
Reject (H_0)	Type I Error (α)	Correct Decision $\rightarrow$ Power ($1 - \beta$)
Fail to Reject (H_0)	Correct Decision	Type II Error (β)



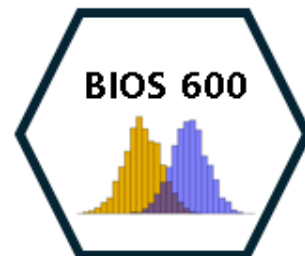
n , detectable difference, variance, and α

How do these four considerations affect power?

- ▶ When we design a study, it is not enough to know we have a small probability of rejecting H_0 when it is in fact true (i.e. setting $\alpha = 0.05$).
- ▶ We want to know we have a large probability of rejecting the null when it is false. Practically speaking, **power less than 80% is typically considered insufficient to warrant a study.**

↪ $1 - \beta = \text{power} = .80$

$\beta = \text{Type 2 error} = .20$



Power

α = typically .05 = rejecting H_0 when H_0 true

α = .01 = less likely to reject H_0 when H_0 true
(and also in general)

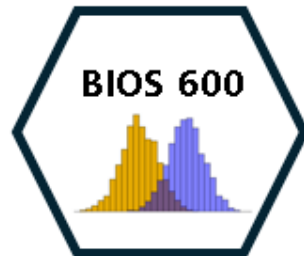
- ▶ Recall: In our medical diagnostics lecture, we saw that there is a tradeoff between Type 1 and Type 2 error. Lowering α makes it harder to reject the null hypothesis, which usually **reduces power**—we're less likely to catch real effects (true positives). $\Downarrow$ **reject H_0 less**

Do not
do this!

* $\uparrow \alpha \Rightarrow \uparrow \text{Power}$

when actually
False
(lower power)

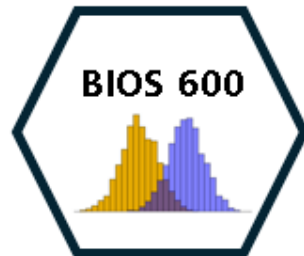
- ▶ Increasing power by tolerating more Type I errors is **not acceptable**. Therefore, we can increase power by
- ▶ Considering **larger deviations from the null** (need to think about clinical/practical importance)
- ▶ Increasing n (good, though not always affordable)



Interactive visualization

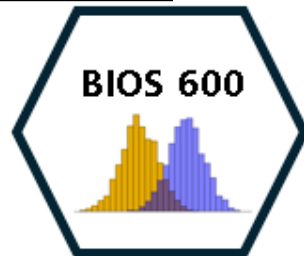
<http://rpsychologist.com/d3/NHST/>

Let's check out this interactive visualization to see how this all works for a one-sample z-test.



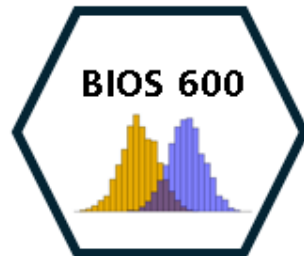
Power Depends On...

Factor	Change	Effect on Power	Intuition
Sample size (n)	↑	↑ Increases	More data → smaller SE → easier to detect true effects
Variability (σ^2)	↑	↓ Decreases	More noise → harder to detect signal
Effect size (δ)	↑	↑ Increases	Larger true difference → easier to detect
Significance level (α)	↑ ↓	↑ Increases ↓	More lenient threshold → higher chance to reject H_0



Understanding power

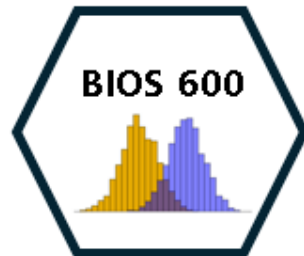
- ▶ Holding n , variance, and α constant, we have **more power** to detect a **larger difference** compared to a smaller one.
- ▶ Holding variance, effect size, and α constant, **increasing sample size (n)** increases power — with more data, we're more likely to detect true effects.



Understanding Power - Your turn!

💡 Circle One

- ▶ Holding n , effect size, and α constant, **(higher/lower)** variance decreases power because the signal is harder to detect within the noise.
- ▶ Holding n , variance, and effect size constant, **(increasing/decreasing)** α increases power — we're more willing to reject the null, so true effects are easier to detect.



What is this deviation?

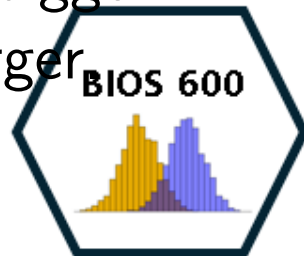
When we calculate power, we need to know the **minimum difference** from the null mean μ_0 that we wish to detect. We may set up our hypotheses as

$$H_0 : \mu = \mu_0$$

$$H_1 : \mu \neq \mu_0$$

delta

but need to specify a **minimum detectable difference**, often called $\delta = \mu_1 - \mu_0$ such that we reject H_0 with a certain power (usually 80% or 90%) when in fact $\mu = \mu_1$. We will need a bigger sample size when δ is small, and fewer subjects when δ is larger.

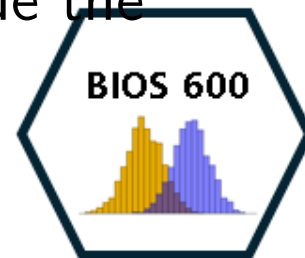


Sample size for two-sided one-sample test of mean

- ▶ Power and sample size are interrelated.
- ▶ Ideally, when we plan a study, we have a prespecified idea of the minimum difference we want to detect, $\delta = \mu_1 - \mu_0$, the standard deviation, σ , and the power, $1 - \beta$, we'd like to have to detect it. In that case, it is straightforward to calculate the required sample size (here given for a one-sample test):

$$n = \left[\frac{\sigma(z_{1-\alpha/2}^* + z_{1-\beta}^*)}{\mu_1 - \mu_0} \right]^2$$

- ▶ This can be derived from looking at the rejection regions when H_0 and H_1 are true, though the details are outside the scope of this class.
- ▶ You can easily calculate power and sample size in R.



Case study: ultra low dose contraception

- ▶ Suppose you want to ensure a new manufacturer of birth control pills provides the correct dosage of $0.02 \mu\text{g}$ estrogen.
- ▶ How many pills do you need to sample in a shipment in order to ensure 80% power to detect a difference of 10% at $\alpha=0.05$, assuming $\sigma=0.008$? (10% of 0.02 is 0.002)

$$n = \left[\frac{\sigma(z_{1-\alpha/2}^* + z_{1-\beta}^*)}{\mu_1 - \mu_0} \right]^2$$

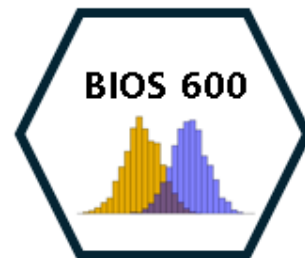
$$\sigma = .008$$

$$\mu_1 - \mu_0 = .002$$

$$\alpha = .05$$

$$1 - \beta = .80$$

What are $z_{1-\alpha/2}^*$ and $z_{1-\beta}^*$? These are the $1 - \alpha$ and $1 - \beta$ quantile cutoffs of a Z-distribution.



Z cutoffs in R

► $\alpha = 0.05$

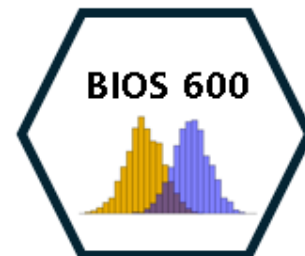
► Power = 80%, so Type 2 error rate is $1 - .80 = .20 = \beta$.

```
alpha <- 0.05 # type 1 error rate  
beta <- .20 # type 2 error rate  
qnorm(1-alpha/2); qnorm(1-beta)
```

[1] 1.959964 ←

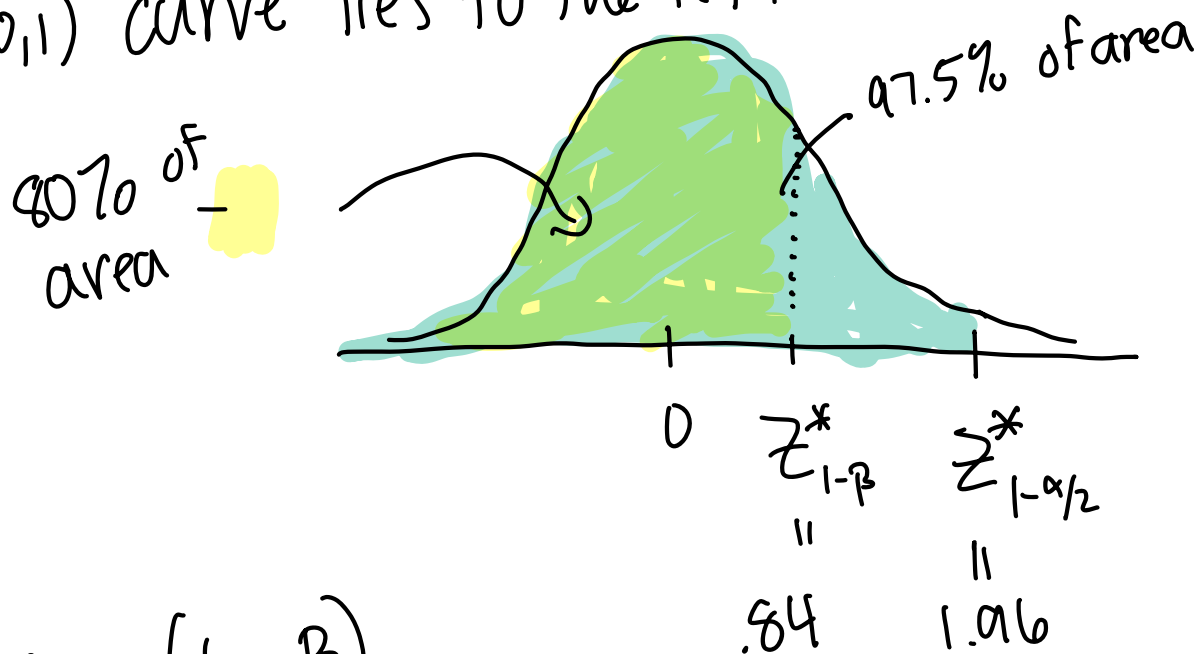
[1] 0.8416212

$$1 - \alpha/2 = 97.5$$



$$qnorm(1 - \alpha/2) = qnorm(.075)$$

What's the cutoff where 97.5% of area under $N(0,1)$ curve lies to the left?



$$qnorm(1 - \beta)$$

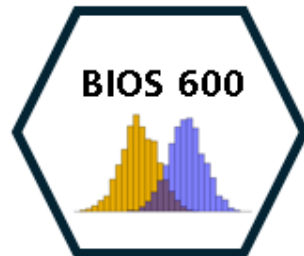
$qnorm(.80)$: what's the cutoff where 80% of area under $N(0,1)$ curve lies to the left?

Plugging in our values

$$= \left[\frac{.008(1.96+0.84)}{0.002} \right]^2$$
$$= 125.44$$

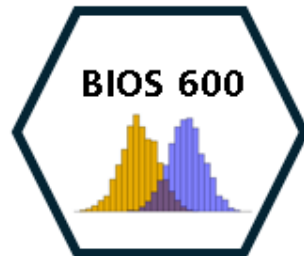
Using a normal approximation, we would need 126 pills sampled to detect the desired difference of 10%, given $\sigma = 0.008$, at 80% power.

- ▶ To be conservative, always **round up for sample size calculations.**



We use the t -distribution in practice

- ▶ In the example above, we used the z -distribution. However, we use the t distribution in practice instead of the z .
- ▶ This is because the population standard deviation (σ) is **rarely known**, so we estimate it from the sample.
- ▶ The t -distribution accounts for this extra uncertainty.
- ▶ When sample sizes are **small or moderate**, using the t -distribution provides more accurate power estimates, since it better reflects the variability from estimating σ .



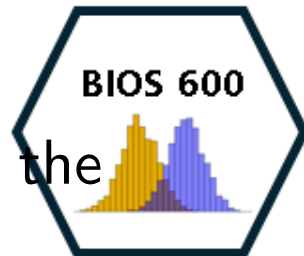
In R

```
power.t.test(delta = 0.002,  
             sd = 0.008,  
             sig.level = 0.05,  
             power = 0.8,  
             type = "one.sample")
```

One-sample t test power calculation

```
          n = 127.5161  
      delta = 0.002  
        sd = 0.008  
sig.level = 0.05  
    power = 0.8  
alternative = two.sided
```

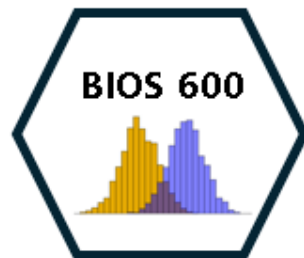
Using the t-distribution, we need 128 pills sampled to detect the desired difference.



Example: Power Calculation

- ▶ Researchers are testing whether a new low-sodium diet reduces **systolic blood pressure (SBP)** compared to a control diet.
- ▶ Previous studies suggest an average reduction of **8 mmHg** with a standard deviation of **10 mmHg**. $\approx f$
- ▶ They plan to recruit **20 participants per group** and use a **two-sided t-test** at the 0.05 level.

Question: What statistical *power* will they have to detect this difference?



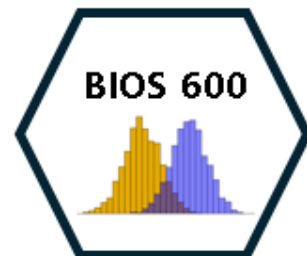
In R

```
# Power calculation for two-sample t-test  
power.t.test(n = 20,  
             delta = 8,  
             sd = 10,  
             sig.level = 0.05,  
             type = "two.sample",  
             alternative = "two.sided")
```

Two-sample t test power calculation

```
          n = 20  
        delta = 8  
          sd = 10  
    sig.level = 0.05  
      power = 0.6933994  
alternative = two.sided
```

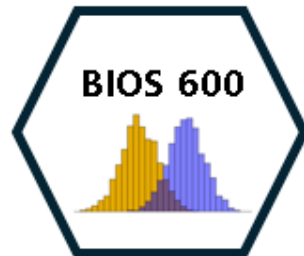
NOTE: n is number in *each* group



Interpretation

Interpretation: With 20 participants per group, the study has about 69% power to detect an 8 mmHg difference.

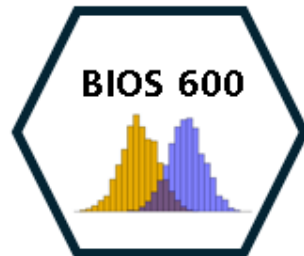
- ▶ In other words, if the true effect exists, there's roughly a 7-in-10 chance we'll correctly reject the null hypothesis.



Example 2: Sample size calculation

- ▶ Suppose we want to test whether a new medication lowers LDL cholesterol compared to a placebo.
- ▶ From pilot data, we expect a difference of 15 mg/dL with a standard deviation of 20 mg/dL.
- ▶ We want 90% power at a 0.05 significance level for a two-sided test.

Question: How many participants per group are needed?



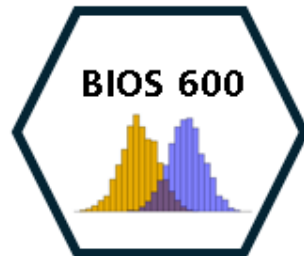
In R

```
power.t.test(power = 0.90,  
             delta = 15,  
             sd = 20,  
             sig.level = 0.05,  
             type = "two.sample",  
             alternative = "two.sided")
```

Two-sample t test power calculation

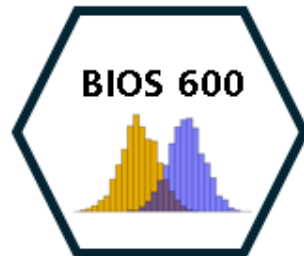
```
      n = 38.34601  
delta = 15  
      sd = 20  
sig.level = 0.05  
      power = 0.9  
alternative = two.sided
```

NOTE: n is number in *each* group



Interpretation

39 individuals per group are needed (rounding up) to achieve 90% power to detect a 15 mg/dL difference in LDL cholesterol between treatments.



Application Exercise

9:05

· 1a ✓

· 2a ✓

· 2b

· Statement ✓

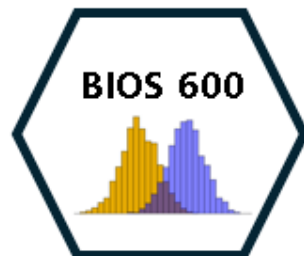
· Statement

💡 AE 04

Head to Canvas and begin working on Application Exercise (AE) 04: Power and Sample Size Calculation.

▶ AE 04 is due **Monday 10/27 at 11:59pm.**

▶ (Turn in a PDF on Canvas.)



Next Class

- ▶ Comparing two continuous variables

